

# Math 189 Homework 1

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## Problem 1

Part a:

```
library(tidyverse)
load("ames.rda")

#Nominal: MS.Zoning

table(ames$MS.Zoning)
```

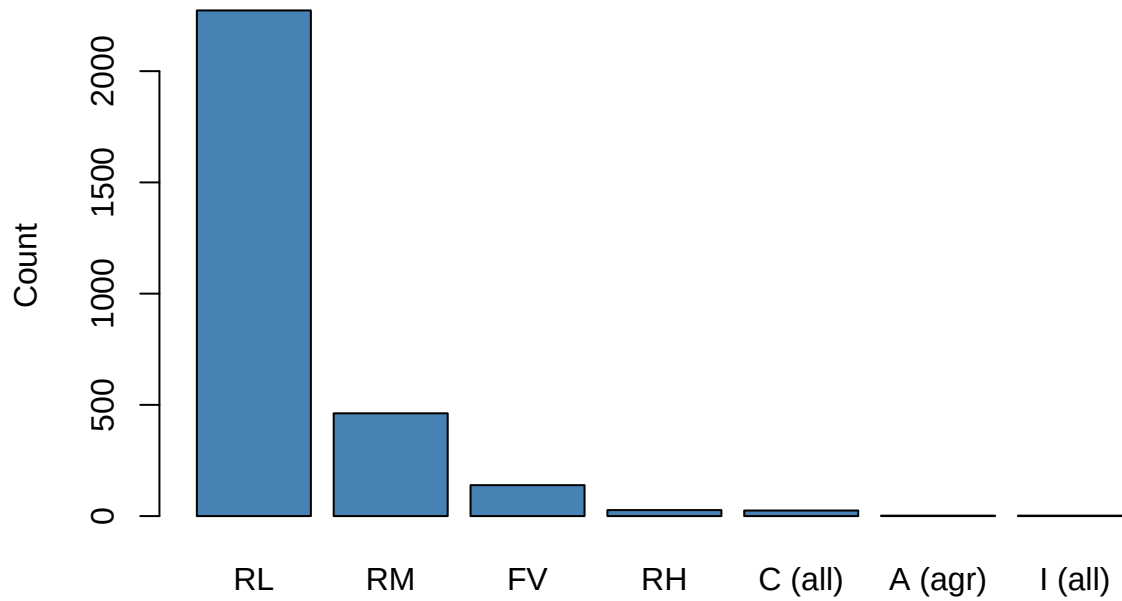
```
##
## A (agr) C (all)      FV I (all)      RH      RL      RM
##      2      25     139      2      27     2273     462
```

```
prop.table(table(ames$MS.Zoning))
```

```
##
##      A (agr)      C (all)      FV      I (all)      RH      RL
## 0.0006825939 0.0085324232 0.0474402730 0.0006825939 0.0092150171 0.7757679181
##      RM
## 0.1576791809
```

```
barplot(sort(table(ames$MS.Zoning), decreasing = TRUE),
        main = "Zoning Classification (Nominal)",
        ylab = "Count", col = "steelblue")
```

## Zoning Classification (Nominal)



*#Ordinal.Exterior.Quality*

```
table(ames$Exter.Qual)
```

```
##
##   Ex   Fa   Gd   TA
## 107   35  989 1799
```

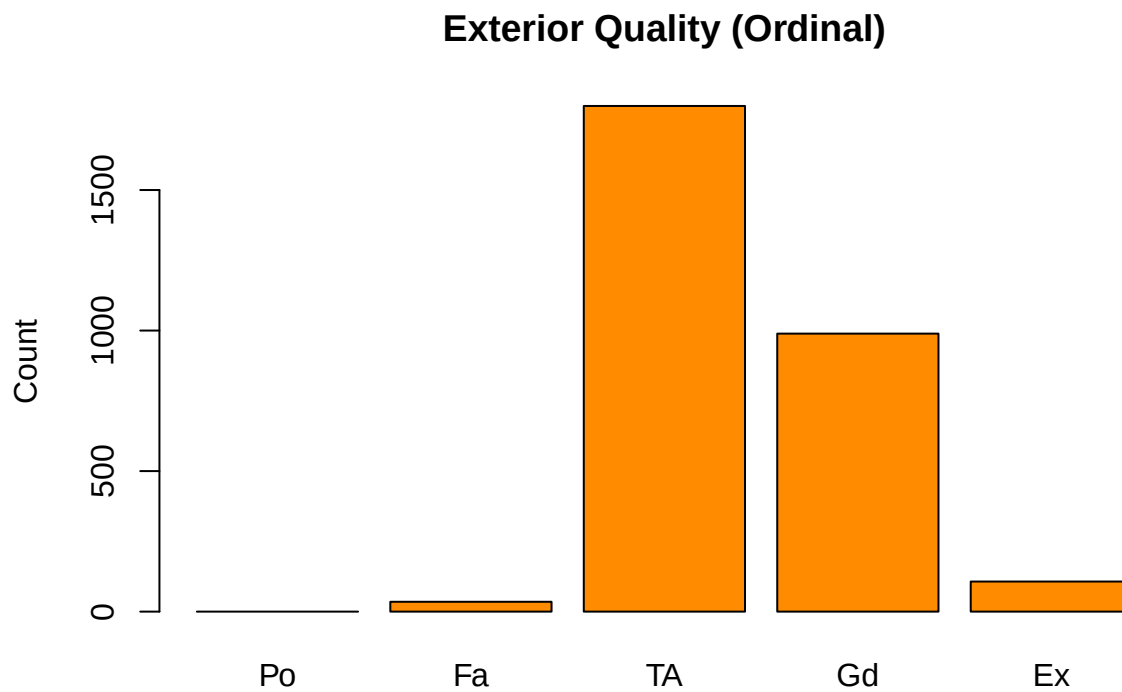
```
ames$Exter.Qual <- factor(ames$Exter.Qual,
                          levels = c("Po", "Fa", "TA", "Gd", "Ex"),
                          ordered = TRUE)
table(ames$Exter.Qual)
```

```
##
##   Po   Fa   TA   Gd   Ex
##    0   35 1799  989 107
```

```
summary(ames$Exter.Qual)
```

```
##   Po   Fa   TA   Gd   Ex
##    0   35 1799  989 107
```

```
barplot(table(ames$Exter.Qual),
        main = "Exterior Quality (Ordinal)",
        ylab = "Count", col = "darkorange")
```



*#Discrete numerical: Bedroom.AbvGr*

```
summary(ames$Bedroom.AbvGr)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##  0.000   2.000   3.000   2.854   3.000   8.000
```

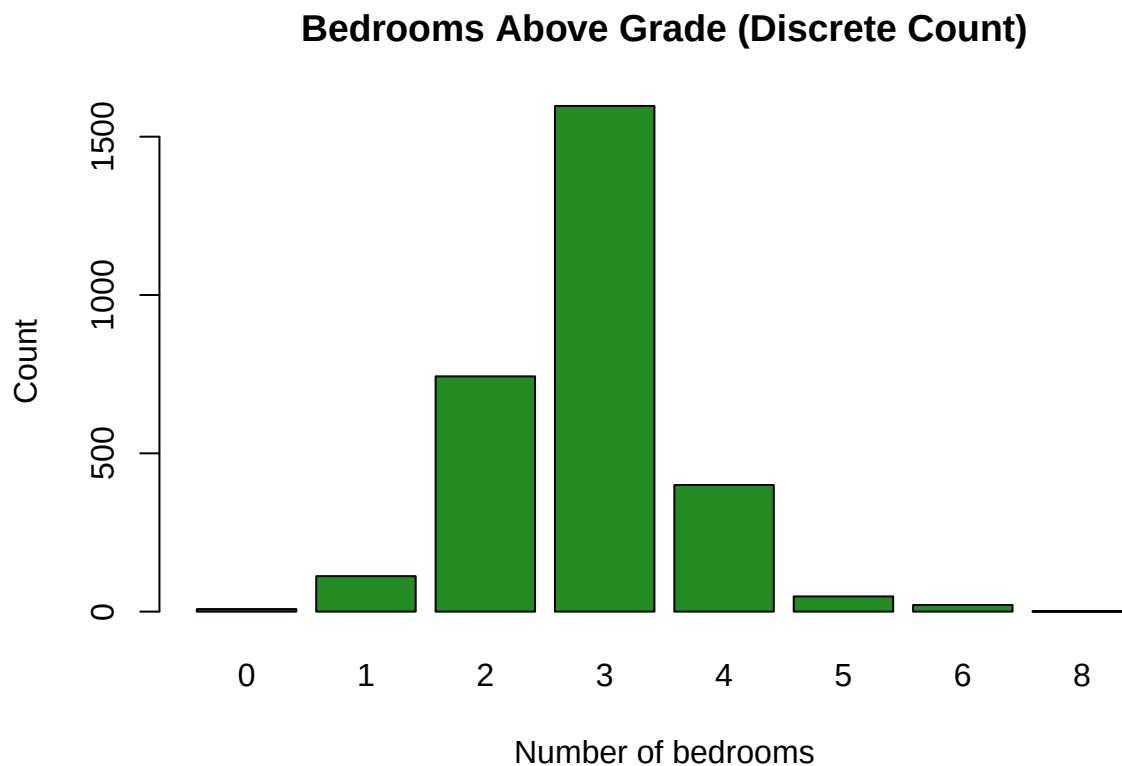
```
sd(ames$Bedroom.AbvGr)
```

```
## [1] 0.8277311
```

```
table(ames$Bedroom.AbvGr)
```

```
##
##      0      1      2      3      4      5      6      8
##      8    112    743   1597    400     48     21     1
```

```
barplot(table(ames$Bedroom.AbvGr),
        main = "Bedrooms Above Grade (Discrete Count)",
        xlab = "Number of bedrooms", ylab = "Count", col = "forestgreen")
```



*#Continuous numerical: area in square feet*

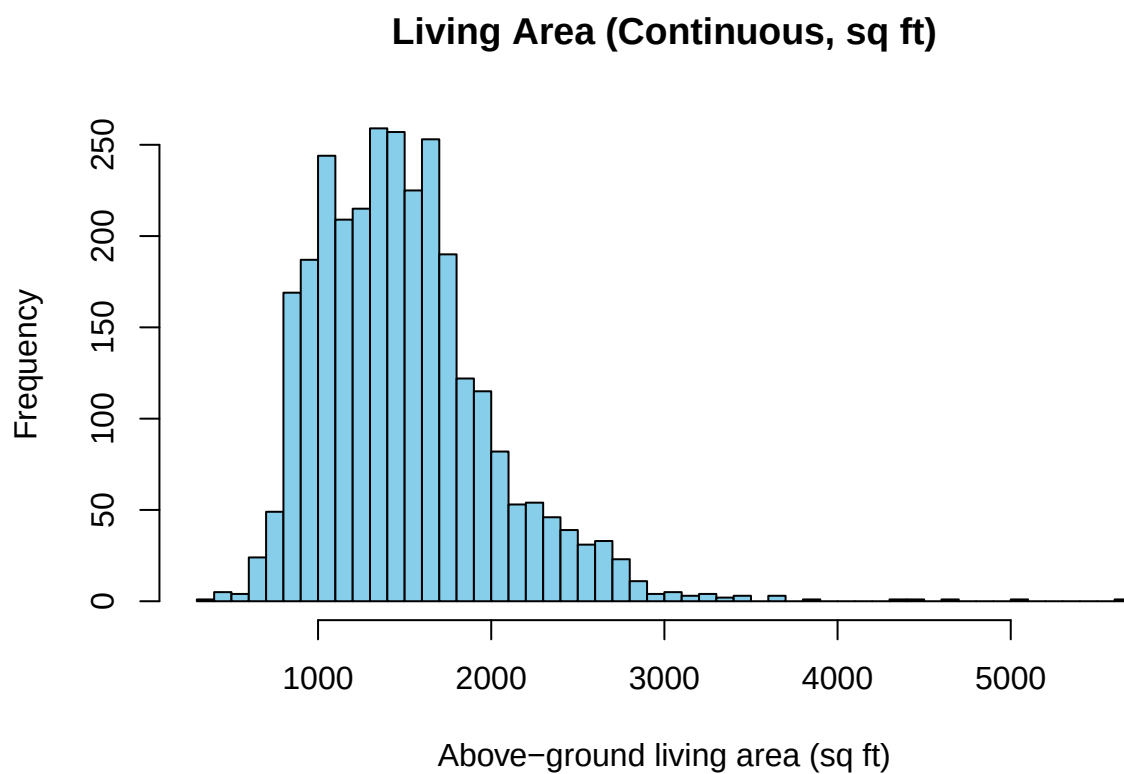
```
summary(ames$area)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      334   1126   1442   1500   1743   5642
```

```
sd(ames$area)
```

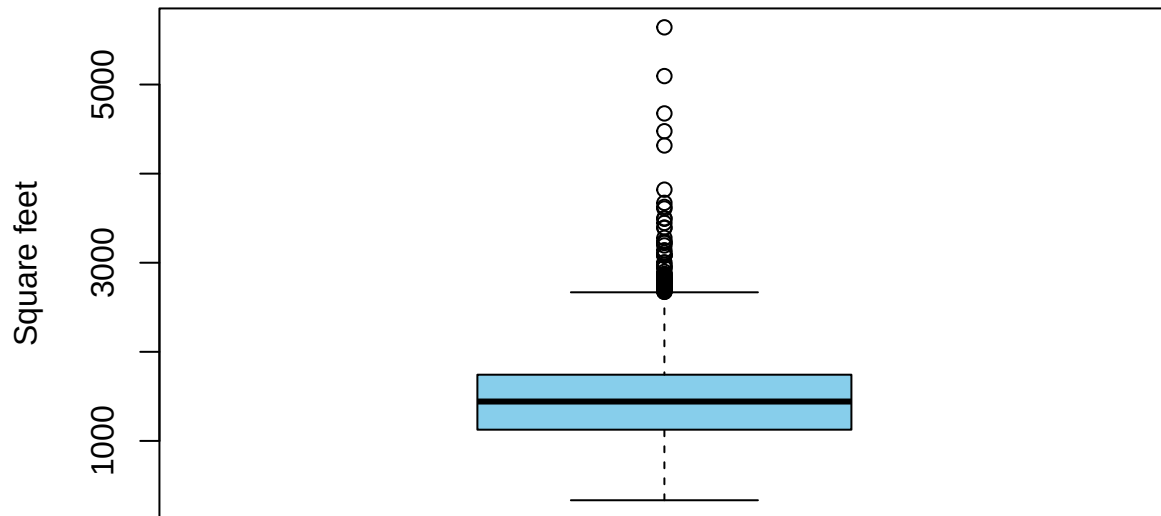
```
## [1] 505.5089
```

```
hist(ames$area, breaks = 40,
     main = "Living Area (Continuous, sq ft)",
     xlab = "Above-ground living area (sq ft)", col = "skyblue")
```



```
boxplot(ames$area,  
        main = "Living Area (Continuous, sq ft)",  
        ylab = "Square feet", col = "skyblue")
```

## Living Area (Continuous, sq ft)



## Observations: \* S.Zoning (nominal variable) — Dominated by “RL” (residential low density): 2273/2930  $\approx 78\%$ , followed by “RM” (residential medium density,  $\sim 16\%$ ). Other categories (agricultural, commercial, industrial, floating village, high density) are rare, some with only 1–2 homes in the sample. Bar chart shows distribution clearly, and makes clear in some later graphs that A (agr) and I (industrial) are excluded from analysis due to small n.

- Exter.Qual (ordinal variable) — Most homes fall in “TA” (typical/average,  $n=1799$ ) or “Gd” (good,  $n=989$ ); very few are “Fa” and none are “Po” in this sample. The ordered bar chart preserves the Po→Ex ordinality sequence, makes chart easy to read, and shows the distribution of quality across the sample.
- Bedroom.AbvGr (discrete count variable) — Mean  $\approx 2.85$  bedrooms, median = 3, SD  $\approx 0.83$ . Almost symmetric and unimodal, most datapoints at 2–4 bedrooms, with a long thin right tail.
- area (continuous variable) — Mean  $\approx 1500$  sq ft, median = 1442, SD  $\approx 506$ , right-skewed (mean  $>$  median) with some large outlier homes. Histogram shows skew clearly, and boxplot flag several high-end outliers beyond the whiskers.

### Part b:

```
table(ames$MS.Zoning, ames$Exter.Qual)
```

```
##
##           Po   Fa   TA   Gd   Ex
##  A (agr)    0    1    1    0    0
```

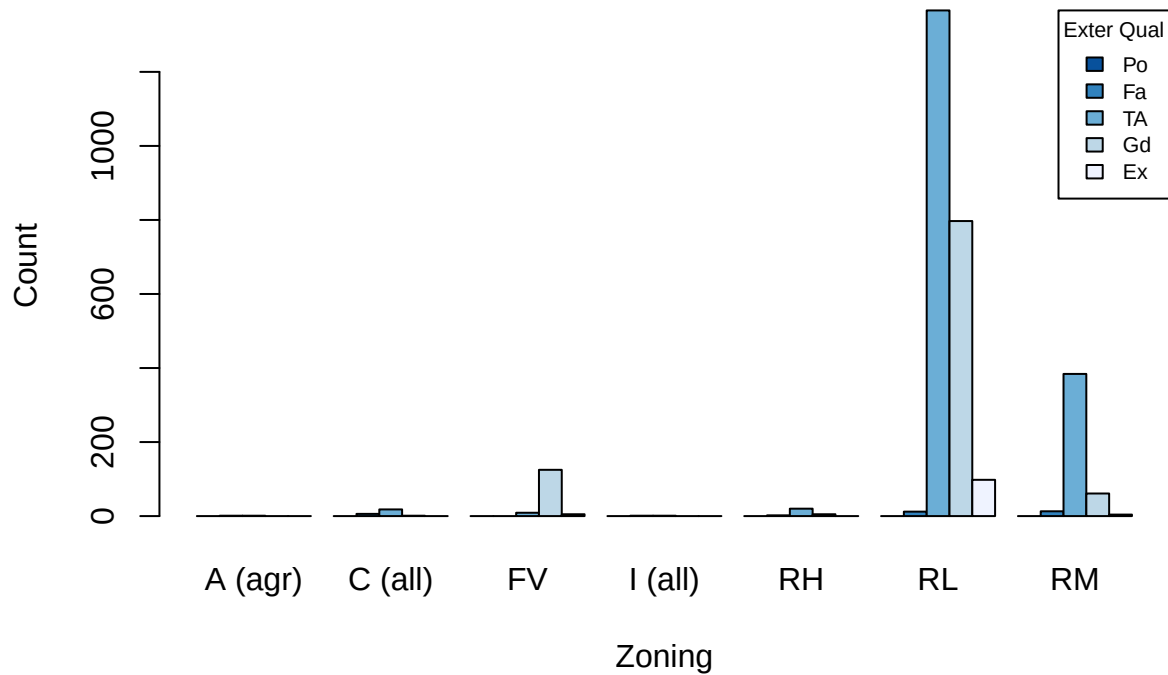
```
## C (all)    0    6   18    1    0
## FV         0    0    9  125    5
## I (all)   0    1    1    0    0
## RH        0    2   20    5    0
## RL        0   12 1366   797   98
## RM        0   13  384    61    4
```

```
# proportions within each zoning class
prop.table(table(ames$MS.Zoning, ames$Exter.Qual), margin = 1)
```

```
##
##              Po              Fa              TA              Gd              Ex
## A (agr) 0.000000000 0.500000000 0.500000000 0.000000000 0.000000000
## C (all) 0.000000000 0.240000000 0.720000000 0.040000000 0.000000000
## FV      0.000000000 0.000000000 0.064748201 0.899280576 0.035971223
## I (all) 0.000000000 0.500000000 0.500000000 0.000000000 0.000000000
## RH      0.000000000 0.074074074 0.740740741 0.185185185 0.000000000
## RL      0.000000000 0.005279366 0.600967884 0.350637923 0.043114826
## RM      0.000000000 0.028138528 0.831168831 0.132034632 0.008658009
```

```
# grouped barplot
zoning_qual_tab <- table(ames$Exter.Qual, ames$MS.Zoning)
barplot(zoning_qual_tab,
        beside = TRUE,
        legend = rownames(zoning_qual_tab),
        col = c("#08519c", "#3182bd", "#6baed6", "#bdd7e7", "#eff3ff"),
        main = "Exterior Quality by Zoning Classification",
        xlab = "Zoning", ylab = "Count",
        args.legend = list(x = "topright", cex = 0.7, title = "Exter Qual"))
```

## Exterior Quality by Zoning Classification



```
library(dplyr)
library(ggplot2)
library(forcats)

ames %>%
  filter(!MS.Zoning %in% c("A (agr)", "I (all)")) %>%
  mutate(
    MS.Zoning = fct_infreq(MS.Zoning),
    Exter.Qual = factor(Exter.Qual, levels = c("Fa", "TA", "Gd", "Ex"))
  ) %>%
  ggplot(aes(x = MS.Zoning, fill = Exter.Qual)) +
  geom_bar(position = "fill", width = 0.7) +
  scale_y_continuous(labels = scales::percent, expand = c(0, 0)) +
  scale_fill_brewer(palette = "Set2", name = "Exterior Quality") +
  coord_flip() +
  labs(
    title = "Exterior Quality Distribution Across Zoning Types",
    x = NULL,
    y = "Proportion of Homes",
    caption = "RH and C (all) have small n; A (agr) and I (all) excluded, n<=2"
  ) +
  theme_minimal(base_size = 14) +
  theme(
    plot.title = element_text(face = "bold", size = 16, hjust = 0),
    plot.subtitle = element_text(color = "grey40", size = 11, margin = margin(b = 12)),
    plot.caption = element_text(color = "grey60", size = 9, hjust = 1),
```

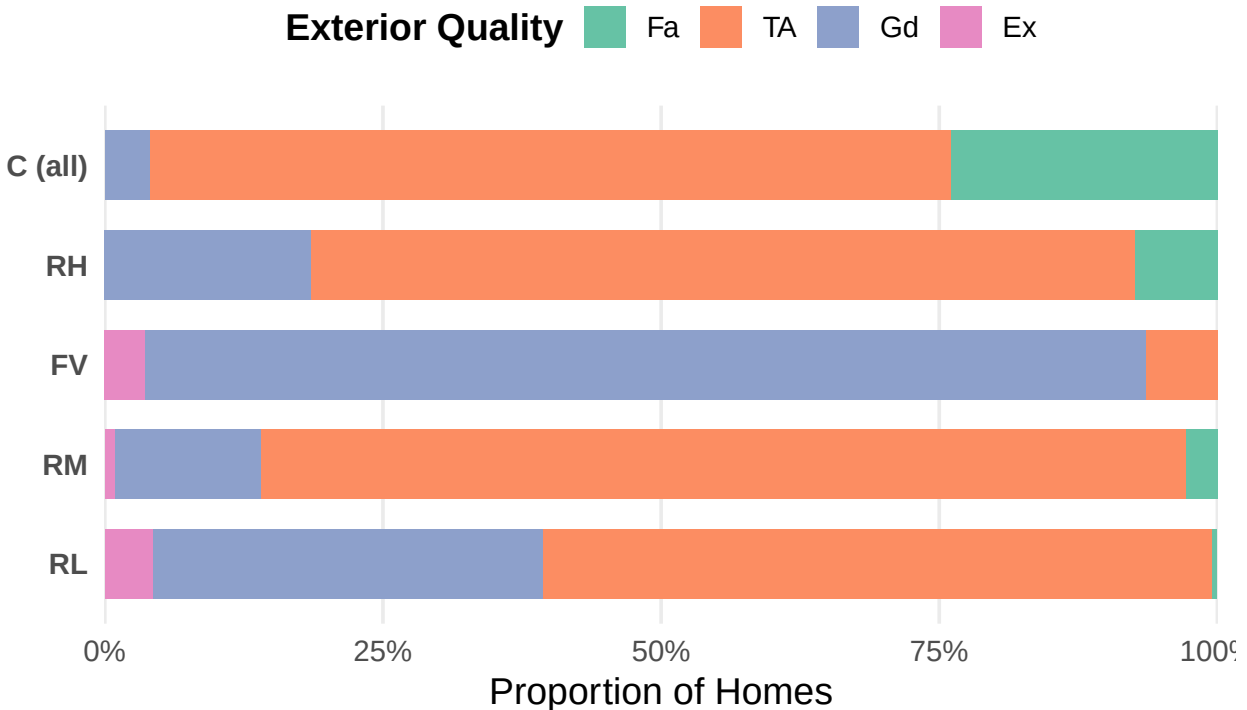


```

legend.position = "top",
legend.title = element_text(face = "bold"),
panel.grid.minor = element_blank(),
panel.grid.major.y = element_blank(),
axis.text.y = element_text(face = "bold")
)

```

## Exterior Quality Distribution Across Zoning Types



RH and C (all) have small n; A (agr) and I (all) excluded, n<=2

## Observations: \*Most homes are in residential low-density zoning, with second-most in residential medium. Typical/average exterior quality dominates across all zoning types except for floating villages where good dominates. Very few homes in agricultural or industrial zoning (<=2 for both so excluded from plot). ## c:

```

load("ames.rda")
summary(ames$area)

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      334   1126   1442   1500   1743   5642

```

```

summary(ames$price)

```

```

##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      12789 129500 160000 180796 213500 755000

```

```
cor_val <- cor(ames[["area"]], ames[["price"]], use = "complete.obs")
print(cor_val)
```

```
## [1] 0.7067799
```

```
plot(ames$area, ames$price,
     pch = 19, col = adjustcolor("steelblue", alpha.f = 0.4),
     main = "Price vs Living Area",
     xlab = "Above-ground living area (sq ft)",
     ylab = "Price ($)")
abline(lm(price ~ area, data = ames), col = "red", lwd = 2)
```



## Observations: \* Positive, roughly linear, relationship between living area and price, with some outliers. Some high outliers on right, and also low outliers on far right, much of data clustered in lower left. More variability in price for larger homes, consistent with heteroskedasticity.

part d:

```
library(dplyr)

ames <- ames %>%
  mutate(Exter.Qual = factor(Exter.Qual, levels = c("Fa", "TA", "Gd", "Ex")))
```

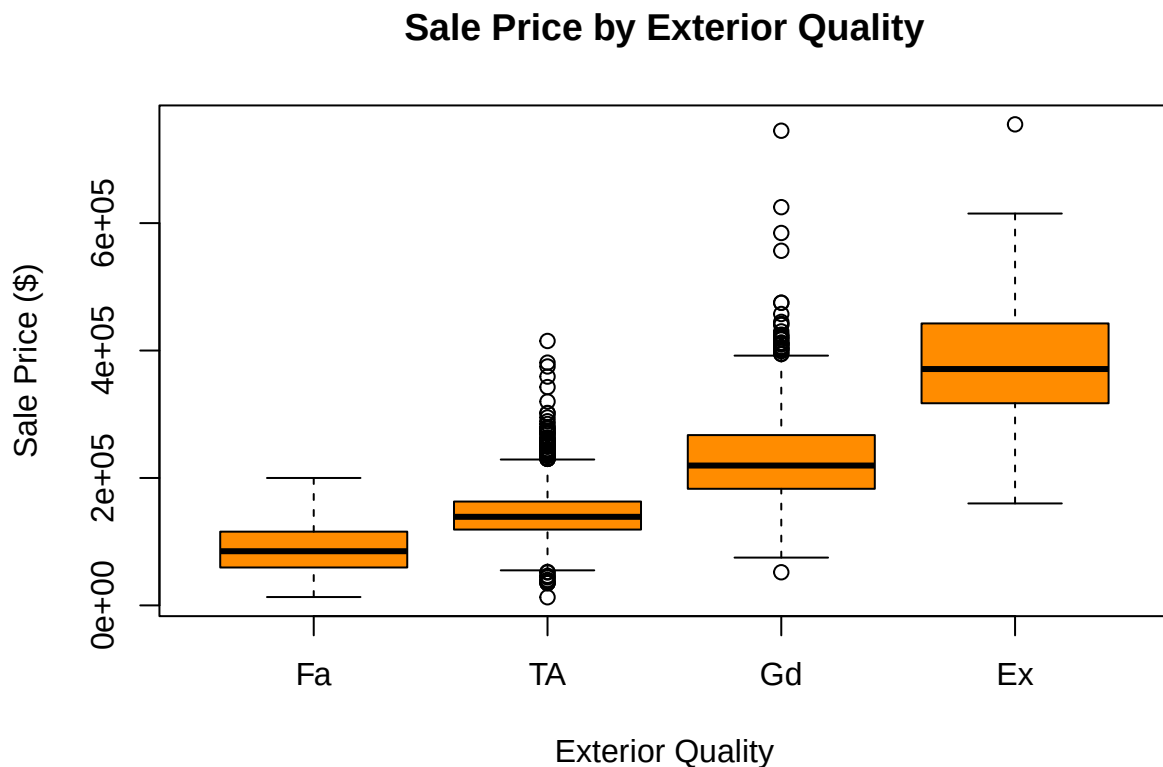
```

ames %>%
  group_by(Exter.Qual) %>%
  summarise(
    Min = min(price, na.rm = TRUE),
    Q1 = quantile(price, 0.25, na.rm = TRUE),
    Median = median(price, na.rm = TRUE),
    Mean = mean(price, na.rm = TRUE),
    Q3 = quantile(price, 0.75, na.rm = TRUE),
    Max = max(price, na.rm = TRUE),
    SD = sd(price, na.rm = TRUE)
  )

## # A tibble: 4 x 8
##   Exter.Qual   Min     Q1 Median   Mean     Q3    Max    SD
##   <fct>       <int> <dbl> <int>   <dbl> <dbl> <int> <dbl>
## 1 Fa         13100 59500 85000  89924. 115750 200000 38014.
## 2 TA         12789 119000 139000 143374. 163000 415000 41504.
## 3 Gd         52000 183000 219500 230756. 267300 745000 70411.
## 4 Ex        160000 317250 370967 377919. 442520. 755000 106988.

boxplot(price ~ Exter.Qual, data = ames,
  col = "darkorange",
  main = "Sale Price by Exterior Quality",
  xlab = "Exterior Quality",
  ylab = "Sale Price ($)")

```



## Observations:

- Median sale price increases steadily as exterior quality rises from “Fa” to “Ex,” and the spread of prices also widens substantially at the top categories —a few “Ex” homes sell for far more than the typical “TA” home, consistent with quality being a strong driver of price. Noticeable outliers exist for “GD” and “TA” , but many houses for both so average still low.

## Problem 2

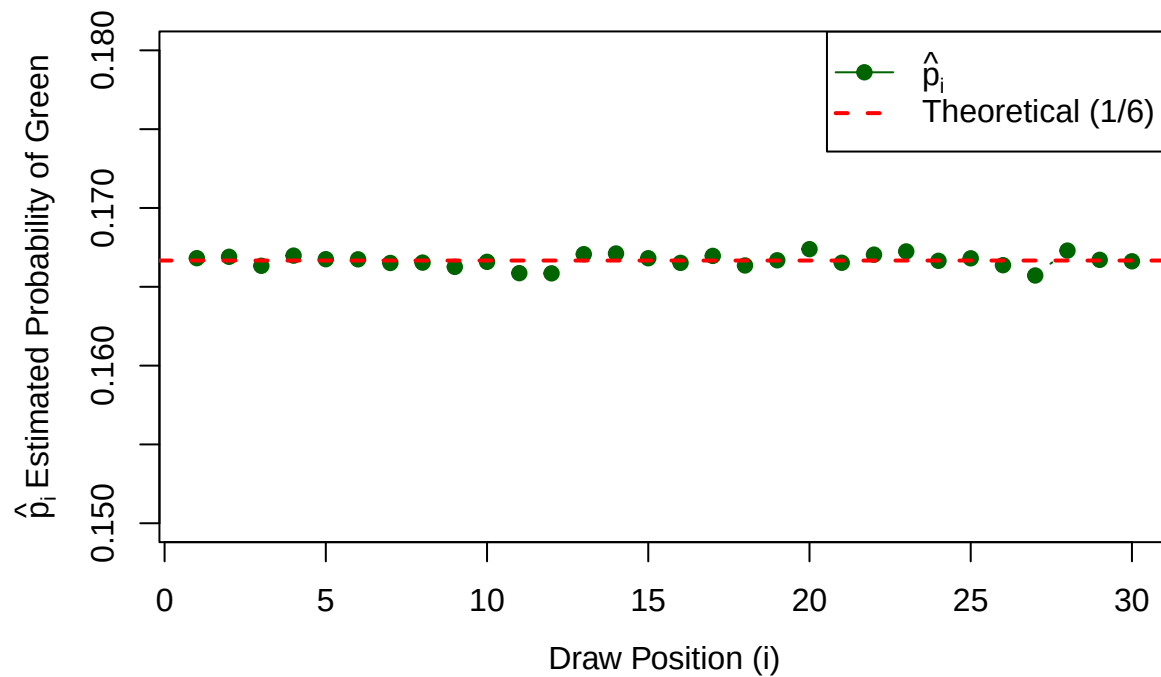
Part a:

```
#c is combine, rep is repeat, this is a 30 element vector
set.seed(42)
urn<-c(rep("red",10), rep("green",5), rep ("blue",15))

n_sims<-1000000
#sample function picks 30 times from urn without replacement
# replicate and n run it 1 million times
#outputs into vector automatically with 30 rows and 1 million columns
# rows are draw position, columns are trial number
simulation_matrix<-replicate(n_sims, sample(urn, size=30, replace=FALSE))
#returns boolean matrix of if green or not
green_check<-(simulation_matrix=="green")
#computes row averages for each draw position, this is a matrix with
#30 rows and 1 column (the mean)
p_hat<-rowMeans(green_check)

#plot
plot(
  1:30, p_hat, type="b", pch = 19, col = "darkgreen",
  xlab="Draw Position (i)", ylab= expression(hat(p)[i] ~"Estimated Probability of Green"),
  main="Estimated Probability of Green by Draw Position",
  ylim = c(0.15, 0.18),
  mgp = c(2.5, 1, 0),
)
#for comparison, line for 1/6
abline(h = 5/30, col = "red", lty = 2, lwd = 2)
legend("topright", legend = c(expression(hat(p)[i]), "Theoretical (1/6)"),
      col = c("darkgreen", "red"), lty = c(1, 2), pch = c(19, NA), lwd = c(1, 2))
```

## Estimated Probability of Green by Draw Position



## Part b:

```
set.seed(42)
urn<-c(rep("red",10), rep("green",5), rep ("blue",15))
n_sims<-1000000

five_balls_alternating_simulation <-function(urn)
{
  draw<-sample(urn, size=5, replace=FALSE)
  alternating_check<-all(draw[-1] != draw[-length(draw)])
  return(alternating_check)
}
results<-replicate(n_sims, five_balls_alternating_simulation(urn))
p_hat<-mean(results)

#output without vector notation in front
cat(sprintf("Estimated Probability of 5 Balls Alternating Colors: %.6f\n", p_hat))
```

## Estimated Probability of 5 Balls Alternating Colors: 0.165609

Part c:

```
set.seed(42)
urn<-c(rep("red",10), rep("green",5), rep ("blue",15))
```

```

n_sims<-1000000

three_balls_sim <-function(urn)
{
  draw<-sample(urn, size=15, replace=FALSE)
  red_count<-sum(draw=="red")
  green_count<-sum(draw=="green")
  blue_count<-sum(draw=="blue")
  return(red_count>=3 && green_count>=3 && blue_count>=3)
}
p_three_balls<-mean(replicate(n_sims, three_balls_sim(urn)))
cat("MC Estimate for (c):", p_three_balls, "\n")

```

```
## MC Estimate for (c): 0.480373
```

### Part d:

```

set.seed(42)
urn<-c(rep("red",10), rep("green",5), rep ("blue",15))
n_sims<-1000000
draws <- replicate(n_sims, sample(urn, size = 10, replace = FALSE))
red_count<-colSums(draws=="red")
green_count<-colSums(draws=="green")
blue_count<-colSums(draws=="blue")
probability_d <- mean(red_count > blue_count)
cat("MC Estimate for (d):", probability_d, "\n")

```

```
## MC Estimate for (d): 0.179479
```

### Part e:

Event I chose: The probability of finding at least 3 non-overlapping 3-ball groups of distinct colors,

(R, G, B in any relative order) within 20 draws without replacement

```

urn <- c(rep("red", 10), rep("green", 5), rep("blue", 15))
n_sims <- 1000000

count_non_overlapping_rgb <- function(urn) {
  draws <- sample(urn, size = 20, replace = FALSE)

  rgb_count <- 0
  i <- 1
  while (i <= length(draws) - 2)
  {
    triplet <- draws[i:(i + 2)]

```

```

    if (length(unique(triplet)) == 3)
    {
        rgb_count <- rgb_count + 1
        i <- i + 3
    }
    else
    {
        i <- i + 1
    }
}
return(rgb_count >= 3)
}

probability <- mean(replicate(n_sims, count_non_overlapping_rgb(urn)))
cat("MC Estimate:", probability, "\n")

## MC Estimate: 0.362007

```

## Problem 3

### Part a:

#### Setup

With  $y = \sqrt{x}$ , so  $x = y^2$  and  $dx = 2y \, dy$ , the limits transform as  $x : 0 \rightarrow 5$  becomes  $y : 0 \rightarrow \sqrt{5}$ :

$$\int_0^5 e^{\sqrt{x}} dx = \int_0^{\sqrt{5}} 2y e^y dy$$

#### Integration by Parts

Let  $u = 2y \implies du = 2 \, dy$ , and  $dv = e^y \, dy \implies v = e^y$ :

$$\int 2y e^y dy = 2y e^y - \int 2e^y dy = 2e^y(y - 1)$$

#### Evaluating the Definite Integral

$$\begin{aligned}
 \left[ 2e^y(y - 1) \right]_0^{\sqrt{5}} &= 2e^{\sqrt{5}}(\sqrt{5} - 1) - 2e^0(0 - 1) \\
 &= 2e^{\sqrt{5}}(\sqrt{5} - 1) + 2 \\
 &\approx 25.130
 \end{aligned}$$

## Part b:

```

f <- function(x)
{
    exp(sqrt(x))
}

```

```
output<- integrate(f, lower=0, upper=5)

cat("Exact Integral:", output$value, "\n")
```

```
## Exact Integral: 25.13047
```

## Part c:

```
set.seed(42)
n_sims <- 1000000
#draws 1 million samples between 0 and 5 randomly
samples<-runif(n_sims,min=0, max=5)
# First integrates all samples
# Second calculates average height above all sampled points
# Multiplies by width of integration interval
monte_carlo_output <- 5*mean(exp(sqrt(samples)))
cat("MC Estimate:", monte_carlo_output, "\n")
```

```
## MC Estimate: 25.13573
```

## Problem 4

```
set.seed(42)

library(tidyverse)

n_values=c(10,20,30,40,50,100)
p_values=c(0.01, 0.02, 0.05, 0.10, 0.20, 0.50)

dataframe_list<-list()
for(n in n_values)
{
  for(p in p_values)
  {
    k<-0:n
    binom_probabilities<-dbinom(k, size=n, prob=p)

    lambda<-n*p
    poisson_probabilities<-dpois(k, lambda=lambda)

    mu<- n*p
    sigma<-sqrt(n*p*(1-p))
    normal_probabilities<-dnorm(k, mean=mu, sd=sigma)

    temp_dataframe<-data.frame(
      n = factor(paste0("n = ", n), levels = paste0("n = ", n_values)),
      p = factor(paste0("p = ", p), levels = paste0("p = ", p_values)),
      k=k,
```



```

    Binomial=binom_probabilities,
    Poisson=poisson_probabilities,
    Normal=normal_probabilities
  )
  dataframe_list[[length(dataframe_list) + 1]] <- temp_dataframe
}
}
#combine all dataframes into one
combined_dataframe <- bind_rows(dataframe_list) %>%
  pivot_longer(cols = c("Binomial", "Poisson", "Normal"),
               names_to = "Distribution",
               values_to = "Density")

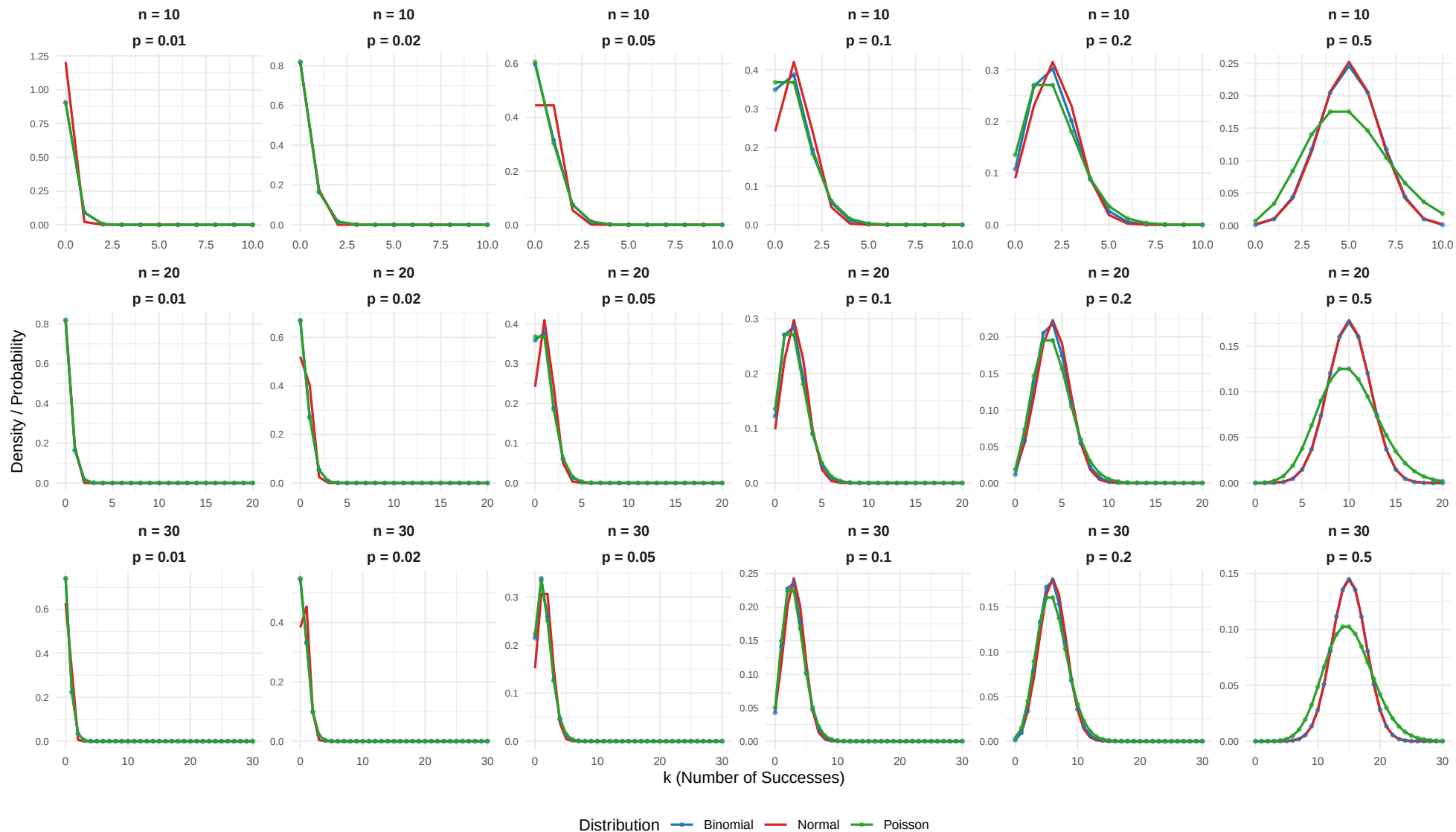
#plot1, small n values (n = 10, 20, 30)
plot1 <- combined_dataframe %>%
  filter(n %in% c("n = 10", "n = 20", "n = 30")) %>%
  ggplot(aes(x = k, y = Density, color = Distribution)) +
  geom_line(linewidth = 0.7) +
  geom_point(data = . %>% filter(Distribution != "Normal"), size = 0.8, alpha = 0.7) +
  facet_wrap(n ~ p, scales = "free", ncol = 6) +
  scale_color_manual(values = c("Binomial" = "#1F77B4",
                                "Poisson" = "#2CA02C", "Normal" = "#D62728")) +
  labs(
    title = "Comparison of Binomial Distribution vs. Approximations
            (n = 10, 20, 30)",
    x = "k (Number of Successes)",
    y = "Density / Probability",
    color = "Distribution"
  ) +
  theme_minimal(base_size = 10) +
  theme(
    strip.text = element_text(face = "bold", size = 8.5),
    axis.text.x = element_text(size = 7),
    axis.text.y = element_text(size = 6),
    legend.position = "bottom"
  )

# Plot 2: Large n values (n = 40, 50, 100)
plot2 <- combined_dataframe %>%
  filter(n %in% c("n = 40", "n = 50", "n = 100")) %>%
  ggplot(aes(x = k, y = Density, color = Distribution)) +
  geom_line(linewidth = 0.7) +
  geom_point(data = . %>% filter(Distribution != "Normal"), size = 0.8, alpha = 0.7) +
  facet_wrap(n ~ p, scales = "free", ncol = 6) +
  scale_color_manual(values = c("Binomial" = "#1F77B4",
                                "Poisson" = "#2CA02C", "Normal" = "#D62728")) +
  labs(
    title = "Comparison of Binomial Distribution vs. Approximations
            (n = 40, 50, 100)",
    x = "k (Number of Successes)",
    y = "Density / Probability",

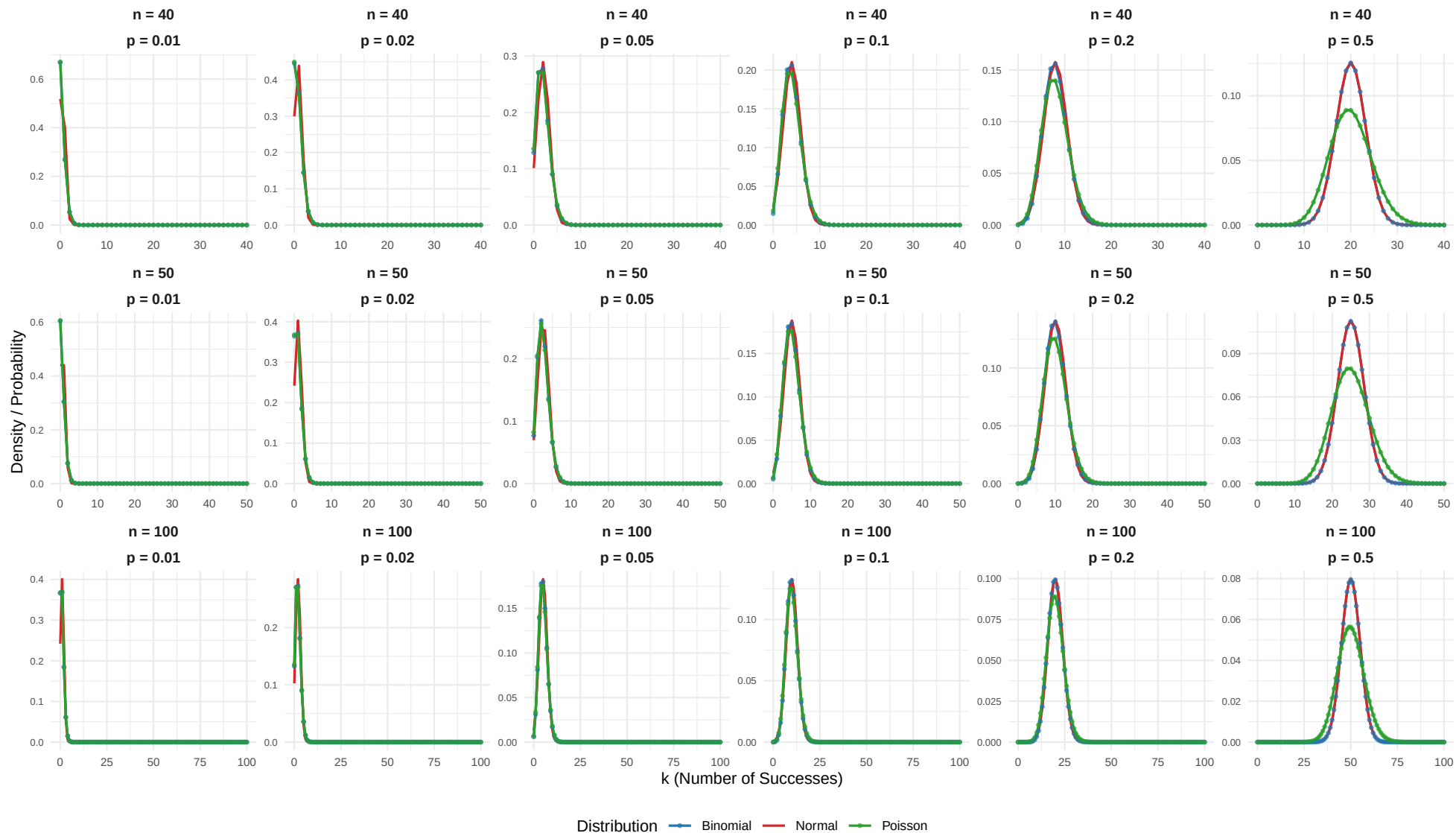
```

```
    color = "Distribution"
) +
theme_minimal(base_size = 10) +
theme(
  strip.text = element_text(face = "bold", size = 8.5),
  axis.text.x = element_text(size = 7),
  axis.text.y = element_text(size = 6),
  legend.position = "bottom"
)
```

# Comparison of Binomial Distribution vs. Approximations (n = 10, 20, 30)



# Comparison of Binomial Distribution vs. Approximations (n = 40, 50, 100)



## Observations

### 1. Binomial Distribution

- Represents the exact discrete probability mass function (PMF).
- Highly skewed when  $p$  is near 0, but becomes symmetrical around  $p = 0.50$ .

### 2. Poisson Approximation

- Fits the Binomial PMF exceptionally well when  $p$  is small ( $p \leq 0.05$ ), even for small  $n$ .
- Rapidly diverges from the true Binomial distribution as  $p$  increases.

### 3. Normal Approximation

- Approaches the Binomial distribution as  $n$  increases ( $n \geq 40$ ) and  $p$  is moderate ( $p \approx 0.50$ ), where  $np \geq 5$  and  $n(1 - p) \geq 5$  hold true.
- Fails for small  $n$  or small  $p$ : as a continuous symmetric curve, it assigns density to negative values where the Binomial PMF has zero probability.

### 4. Summary

- Poisson approximates the Binomial distribution for rare events ( $p \ll 1$ ).
- Normal approximates the Binomial distribution for large, balanced samples.